CONOID: Persistent Per-Skill Reputation and Multi-Channel Judge Feedback for LLM-Agent Organizations

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Abstract

Frameworks for multi-agent language-model systems make it cheap to instantiate named agents and have them coordinate, but most evaluations treat each run as disposable: a team is chosen, an answer is produced, and almost no structured evidence survives into the next assignment. We study that missing persistence layer. CONOID maintains a population of agents that carry profiles, tiered memories, signed social edges, and a per-skill Beta-reputation posterior; a structured judge proposes bounded updates to those channels, and the next team is selected against the updated state. Our contribution is empirical as much as architectural, and the main result is conditional on profile quality. On a 691,200-row oracle-backed team-selection benchmark, a fixed static-role assignment is the best non-oracle policy when visible profiles are reliable (oracle regret 0.044), but a reputation-plus-exploration policy is best when profiles are weak (85.2% success vs. 75.5% for static roles; regret 0.076 vs. 0.098) or adversarially noisy (65.9% vs. 44.2%; regret 0.094 vs. 0.146), with seed-paired bootstrap intervals excluding zero. On a 2,880-row live private-evidence benchmark in which an answer is correct only if the selected team collectively holds all evidence, the same policy closes 68–71% of the random-to-oracle selection gap when evidence holders are profile-aligned, while hidden-expert discovery stays near chance. A 1,440-call coordinator replay isolates a second, separable bottleneck: schema-constrained answer aggregation drops accuracy from 0.839 to 0.586, whereas deterministic slot assembly recovers 0.992. The claim is therefore narrow: persistent per-skill reputation with explicit exploration helps team formation when profiles are incomplete or misleading; it is not a universal substitute for good profiles, and rewarding raw social cohesion can hurt.

1 Introduction

Composing many language-model (LLM) workers into a single system is now common practice. AutoGen, MetaGPT, and ChatDev showed that conversation, role specialization, and staged workflows are enough to get agents cooperating on non-trivial tasks (Wu et al., 2023; Hong et al., 2024; Qian et al., 2024), and orchestration layers such as Magentic-One add a ledger-driven planner over a heterogeneous worker pool (Fourney et al., 2024). These frameworks treat agents as essentially stateless. A team is assembled from fixed role labels, it produces an answer, and the run is discarded; nothing about which agent proved reliable at which skill, which collaborators worked well together, or which assumptions turned out wrong is carried into the next assignment. An organization built this way does not get better with use, which is one reason agent pipelines are easy to demonstrate and hard to move into dependable production.

We treat self-improvement as structured state transition rather than free-form post-hoc reflection. CONOID maintains a persistent population of agents; each agent owns four mutable channels — profile hints, tiered

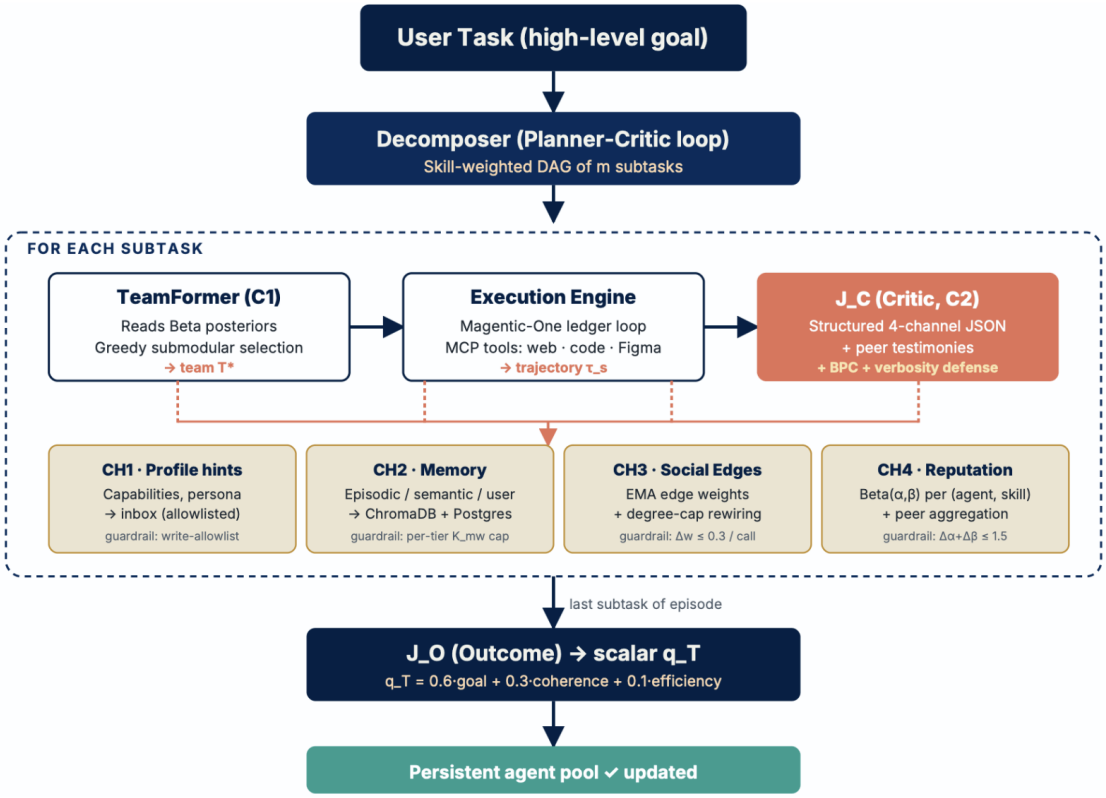


Figure 1: **The CONOID episode loop.** A decomposer turns the user task into a skill-weighted DAG of subtasks through a planner–critic loop. For each subtask, the team former (C1) reads the agents' per-skill Beta posteriors and selects a team by greedy submodular optimization; the team executes inside a Magentic-One-style ledger loop with optional tools and returns a trajectory. A subtask critic J_C (C2) then emits one structured JSON object that updates four channels at once — profile hints, tiered memory, social edges, and per-skill reputation — under per-channel guardrails and the bias defenses of Section 3.3. After the subtask tree finishes, an outcome judge J_O scores the episode and the updated pool becomes the prior for the next task.

memory, signed social edges, and a per-skill Beta-reputation posterior. A task is decomposed into skill-weighted subtasks; for each subtask a team former scores candidate teams against the current state using coverage, reliability, exploration, redundancy, and (optionally) social cohesion; the selected team executes; and a structured judge proposes bounded updates to the four channels, committed by deterministic apply rules. The next task is routed through the changed state, so an estimate of who is reliable at what — with the memories and collaboration history that justify it — accumulates across episodes. Figure 1 shows the loop. We use the name CONOID to distinguish this system from the unrelated "Internet of Agents" collaboration protocol of Chen et al. (2024): our object of study is the persistent state that governs future team formation, not the messaging substrate.

This paper makes two arguments. The first is constructive: a persistent per-skill reputation signal, coupled with explicit exploration, improves team formation when the information a planner would otherwise rely on — declared agent profiles — is weak or misleading. The second is methodological. Evidence about a system like this has to be separated by what each experiment can actually establish. A controlled coordination substrate can show that the state channels and judge protocol are load-bearing; it cannot, on its own, establish live task superiority. An oracle-backed benchmark can measure team-selection quality cleanly because it knows the

ground-truth optimal team; it says nothing about natural-language answer quality. Small live pilots validate instrumentation and expose transfer gaps; they are underpowered for confirmatory claims. We organize the evaluation as an explicit evidence ladder (Section 4) so that each result is read against the claim it supports.

Contributions.

- A persistent LLM-agent organization in which a structured judge updates four auditable state channels — profile hints, tiered memory, signed social edges, and per-skill Beta reputation — through bounded, individually ablatable apply rules (Section 3).
- A reputation-conditioned team former built on per-agent, per-skill Beta posteriors, with exploration driven by posterior variance and a cold-start anti-oligarchy mechanism; the coverage core admits the standard $(1-1/e)$ greedy guarantee (Section 3.2).
- A 691,200-row oracle-backed benchmark that locates the regime where learned reputation helps — weak or adversarial profiles — and a matching negative control where static roles are and should be best (Section 5).
- A live private-evidence benchmark and coordinator replay that separate *evidence-holder selection* from downstream *answer aggregation*, showing the second is a distinct failure mode (Sections 6 and 7).
- An internal-validation suite (channel ablation, anti-oligarchy, adversarial self-promotion, inter-judge agreement) reported as mechanism evidence, not as live task superiority (Section 9).

2 Related Work

LLM-agent organizations. AutoGen, MetaGPT, and ChatDev established practical ways to compose LLM workers through conversation, role specialization, and staged software pipelines (Wu et al., 2023; Hong et al., 2024; Qian et al., 2024); Magentic-One added an orchestrator that plans over and re-assigns a generalist worker pool (Fourney et al., 2024). AgentNet studies decentralized, evolutionary coordination among LLM agents (Yang et al., 2025), and AgentsNet provides a benchmark for coordination and collaborative reasoning that scales to large agent graphs (Grötschla et al., 2025). These systems supply orchestration and communication; they do not maintain a persistent, per-skill, judge-updated reputation that conditions future team formation, which is our focus.

Memory and persistence. Generative Agents introduced durable retrieval scored by recency, importance, and relevance (Park et al., 2023); MemGPT framed memory management as an operating-system problem under a bounded context (Packer et al., 2023); A-MEM and G-Memory added agentic note structures and hierarchical multi-agent memory (Xu et al., 2025; Zhang et al., 2025). CONOID treats memory as necessary but not sufficient: memory records what happened, while reputation is the signal that decides who to select next. Our memory tier follows this prior work; the contribution is its coupling to reputation-gated reads and judge-driven retraction.

Trust and reputation. The Beta reputation system gives a compact posterior over positive and negative evidence (Jøsang and Ismail, 2002); FIRE and the evidential model of Yu and Singh (2002) combine direct experience with peer testimony in open multi-agent systems (Huynh et al., 2006). Recent work shows reputation can avert cooperation collapse in LLM-based multi-agent systems (Ren et al., 2026). We adopt this statistical substrate but make it *per skill* and task-conditioned, and we keep the actor, the peer witnesses, and the judge model separate so that no single party can inflate an agent's standing unilaterally.

Self-improvement. Reflexion improves a single agent through verbal feedback (Shinn et al., 2023); Self-Rewarding LMs and SiriuS improve by fine-tuning on self- or critic-generated preference data (Yuan et al., 2024; Zhao et al., 2025). These require gradient access or collapse agent identity into weights. CONOID freezes model weights and evolves the state *around* each agent, so an agent's identity and accumulated reputation survive model swaps.

LLM-as-judge and evaluator bias. MT-Bench and Chatbot Arena made LLM judging operational (Zheng et al., 2023); G-Eval added rubric-based, probability-weighted scoring (Liu et al., 2023); Agent-as-a-Judge inspects full trajectories rather than only final answers (Zhuge et al., 2024). LLM judges carry position, verbosity, and self-preference biases (Wang et al., 2024; Saito et al., 2023; Dubois et al., 2024; Panickssery et al., 2024). We therefore wire calibration into the judge and, more importantly, keep deterministic labels, oracle regret, evidence ceilings, and parsed-slot assembly as the primary metrics wherever a task admits them, treating the LLM judge score as secondary.

3 The CONOID System

CONOID runs as a closed loop over a persistent agent pool (Fig. 1). A decomposer converts each user task into a skill-weighted DAG of subtasks through a planner–critic loop. For every runnable subtask, the team former (Section 3.2) selects a small team from the pool; the team executes inside a Magentic-One-style ledger loop (Fourney et al., 2024) with optional tool access and returns a trajectory τ_s. Teammates file peer testimony, and a subtask critic J_C inspects τ_s and emits the four-channel update of Section 3.3. After the subtask tree completes, an outcome judge J_O scores the episode with $q_T = 0.6\,\text{goal} + 0.3\,\text{coherence} + 0.1\,\text{efficiency}$ — the scalar we track across episodes — and the updated profiles, memories, edges, and posteriors become the prior for the next task. The rest of this section specifies each component.

3.1 Persistent state

Let $V = \{a_1, \ldots, a_N\}$ be a persistent population and K a fixed canonical skill taxonomy. Free-text skills are snapped to canonical skills by embedding cosine similarity above a threshold $\tau_{\text{snap}} = 0.6$. Each agent a_i carries: (i) profile hints and declared roles; (ii) memories at user, agent, and organization tiers; (iii) directed social edges $w_{ij} \in [0, 1]$; and (iv) a per-skill reputation posterior $\text{Beta}(\alpha_{ik}, \beta_{ik})$. We store $\alpha = \beta = 0$ as the uninformative state and apply a $+1$ Laplace correction at read time, so the reliability estimate and its uncertainty are

$$\rho_{ik} = \frac{\alpha_{ik} + 1}{\alpha_{ik} + \beta_{ik} + 2}, \qquad \sigma_{ik} = \sqrt{\frac{(\alpha_{ik} + 1)(\beta_{ik} + 1)}{(\alpha_{ik} + \beta_{ik} + 2)^2\,(\alpha_{ik} + \beta_{ik} + 3)}}. \tag{1}$$

Reputation is per skill rather than scalar because an agent can be reliable at code review and unreliable at multi-document synthesis. At the start of each episode the posteriors decay geometrically, $\alpha_{ik} \leftarrow \lambda\alpha_{ik}$ and $\beta_{ik} \leftarrow \lambda\beta_{ik}$ with $\lambda = 0.99$, before new evidence is added; this is the discounted-likelihood form of Jøsang and Ismail (2002) and prevents stale evidence from dominating.

3.2 Team formation

For a subtask s with normalized required-skill weights $\mathbf{w}_s = \{w_{sk}\}$, a candidate team $T \subseteq V$ is scored by

$$F(T; s) = C(T; s) + \gamma_{\text{rel}} R(T; s) + \gamma_{\text{exp}} E(T; s) + \gamma_{\text{soc}} S(T) - \gamma_{\text{red}} D(T; s). \tag{2}$$

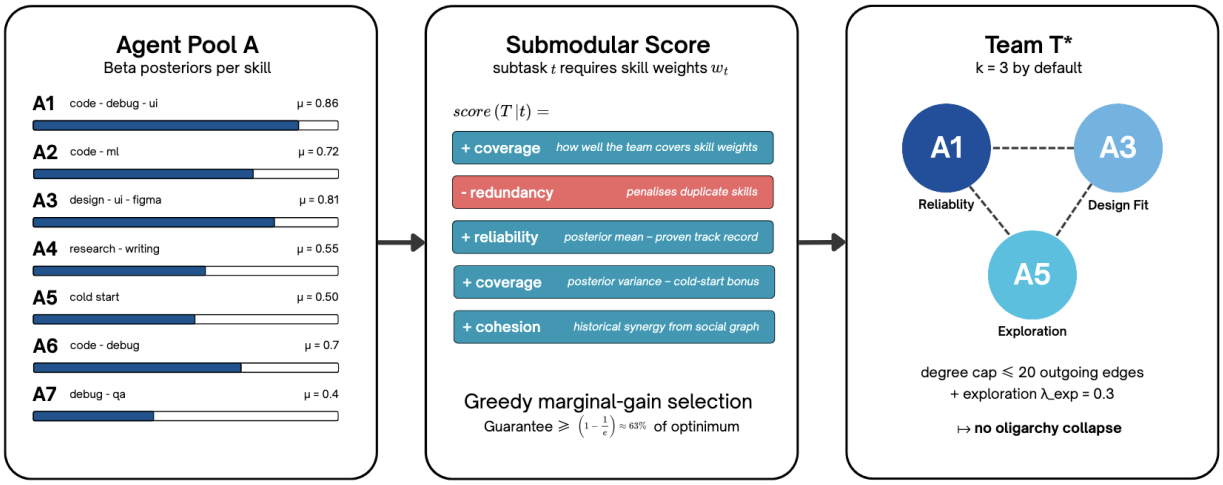


Figure 2: **Reputation-conditioned team formation.** Each candidate agent is summarized by its per-skill Beta-posterior mean. For a subtask with required-skill weights $\mathbf{w}_t$, the five-component submodular objective (Eq. (2)) scores candidate teams, and greedy marginal-gain selection returns a size-m team within $(1-1/e) \approx 0.63$ of the optimum on the coverage core. In the illustrated pool a reliable generalist (A1) and a design specialist (A3) are chosen for coverage and reliability, and a high-uncertainty cold-start agent (A5) is added by the exploration term; the degree cap and exploration weight keep selection from collapsing onto a few agents.

Coverage is a noisy-OR over per-skill reliabilities, which is monotone and submodular; reliability is a per-skill lower-confidence bound aggregated by its best holder; exploration rewards posterior uncertainty; redundancy penalizes duplicated strong coverage; and social cohesion is mean pairwise edge strength:

$$C(T;s) = \sum_k w_{sk}\Big(1 - \textstyle\prod_{i\in T}(1-\rho_{ik})\Big), \qquad R(T;s) = \frac{1}{\sum_k w_{sk}} \sum_k w_{sk} \max_{i\in T}\big(\rho_{ik} - \kappa\sigma_{ik}\big),$$
$$E(T;s) = \frac{1}{\sum_k w_{sk}} \sum_k w_{sk} \sum_{i\in T} \sigma_{ik}, \qquad D(T;s) = \sum_k w_{sk}\Big(\textstyle\sum_{i\in T}\rho_{ik} - C_k(T)\Big), \tag{3}$$

where $C_k(T) = 1 - \prod_{i\in T}(1-\rho_{ik})$ is the per-skill coverage and $S(T) = \mathrm{mean}_{i\neq j\in T}\frac{1}{2}(w_{ij}+w_{ji})$. Teams are built greedily, adding the highest-marginal-gain agent each round and stopping when the gain is non-positive (Fig. 2). The default configuration uses γ_{rel}=1.0, γ_{red}=0.5, γ_{exp}=0.3, γ_{soc}=0.5, κ=0.5, and team size m=3.

Cold-start anti-oligarchy. With every agent at the uninformative prior, all candidates tie and a strict greedy collapses onto whichever agents appear first in iteration order. Two mechanisms prevent this: a per-(episode, subtask) seeded shuffle of the candidate order, and an ϵ-tolerance tied set (ϵ=10^{-3}) from which the winner is drawn uniformly, so strict winners still win cleanly and only genuine near-ties get a coin flip. A complementary degree cap on the edge channel ($d_{\max}$=20 outgoing edges above strength 0.05) keeps the social graph from concentrating.

Proposition 1 (Coverage admits a $(1-1/e)$ guarantee)**.** *The coverage function* $C(T;s) = \sum_k w_{sk}\big(1 - \prod_{i\in T}(1-\rho_{ik})\big)$ *with* $w_{sk} \geq 0$ *is monotone and submodular in* T*. Under a cardinality constraint* $|T| = m$*, greedy selection returns coverage at least* $(1-1/e)$ *of the optimum (Nemhauser et al., 1978).*

Each per-skill term is a weighted coverage function, hence monotone submodular; a non-negative weighted sum preserves both properties, and Nemhauser et al. (1978) applies. The guarantee covers the coverage core only; the reliability, exploration, redundancy, and social terms are bounded perturbations, and we evaluate the full objective empirically.

3.3 Judge update protocol

After a subtask attempt, the subtask critic J_C emits four channel families, each committed by a deterministic, bounded apply rule rather than by free-form rewriting:

- **Reputation.** For each rated (agent, skill) pair the judge's 1–5 Likert rating maps to $\text{obs} = (\text{rating}-1)/4 \in [0,1]$ with an evidence weight $\nu \in \{0.5, 1.0, 1.5\}$, giving $\Delta\alpha = \text{obs} \cdot \nu$ and $\Delta\beta = (1-\text{obs}) \cdot \nu$, clipped per call.
- **Social edges.** Signed deltas drive an EMA of the pairwise edge toward $(\delta+1)/2$ at rate 0.9, subject to the degree cap.
- **Memory.** WRITE/RETRACT/REWRITE operations at the user/agent/org tiers, capped at five writes per call (at most two per tier).
- **Profile hints.** Append-only suggestions to a per-agent inbox (cap fifty, at most three per call); the agent's own reflection decides whether to adopt a hint.

Peer testimony is folded in by Yu–Singh credibility-weighted aggregation: rater j's weight is $c_j \cdot a_t$ where c_j is a persistent credibility ($[0.1, 1.0]$, EMA-updated by agreement with the judge) and $a_t = 1 - |r_j - r_{\text{judge}}|$ is the per-call alignment; the combined judge-plus-peer increment is bounded by $\Delta_{\max} = 1.5$ (Yu and Singh, 2002; Huynh et al., 2006). Three bias defenses are built into the protocol: balanced position calibration (K=2 paired orderings, median with an IQR $\leq$ 1 stability gate for Likert ratings, swap-and-agree for binary and signed channels); length-controlled residualization of the observed rating against within-team log-token counts (α_ℓ=0.15) (Dubois et al., 2024; Saito et al., 2023); and a runtime check that the judge model is disjoint from the agent pool, which we revisit empirically in Section 9.

Policies under test. We compare the default full objective (FULL; all five terms at their default weights) against REP+EXPLORE, the configuration selected by an offline policy sweep: social scoring disabled (γ_{soc}=0, edge channel off) and exploration raised to γ_{exp}=0.60, retaining per-skill reputation and credibility weighting. Baselines are STATIC (a fixed role-to-agent assignment), NO-REP/SKILL-PRIOR (declared-skill matching with no learned posteriors), RANDOM, and oracle references.

4 Evaluation Overview

We evaluate CONOID as an evidence ladder rather than a single leaderboard, and state up front what each block may and may not establish (Table 1). This is the methodological core of the paper: the strongest, fully reproducible evidence (offline oracle regret; live evidence-coverage ceilings) carries the team-selection claim, while the internal-validation suite is reported as mechanism evidence and the live pilots as instrumentation and transfer diagnostics.

Statistics. Offline confidence intervals are seed-paired bootstrap intervals over seed-level means (Efron, 1979); we report seed-paired regret deltas and the fraction of seeds on which a policy wins. Live pilots report point estimates and paired differences only, and are labeled as such.

Table 1: **Evidence ladder.** Each block is read against the claim it supports.

Block	What it tests	What it cannot establish
Offline profile-sensitivity (Section 5)	Whether reputation and exploration lower oracle regret under controlled profile quality.	Natural-language answer quality.
Live private evidence (Sections 6 and 7)	Whether selected teams cover required private evidence, and whether aggregation loses it.	Solving hidden-expert discovery.
Live backend pilots (Section 8)	Whether real channel-isolated episodes run with deterministic scoring.	Confirmatory superiority; samples are small.
Internal validation (Section 9)	Whether channels, anti-oligarchy controls, and bias defenses behave as designed.	Live LLM task superiority.

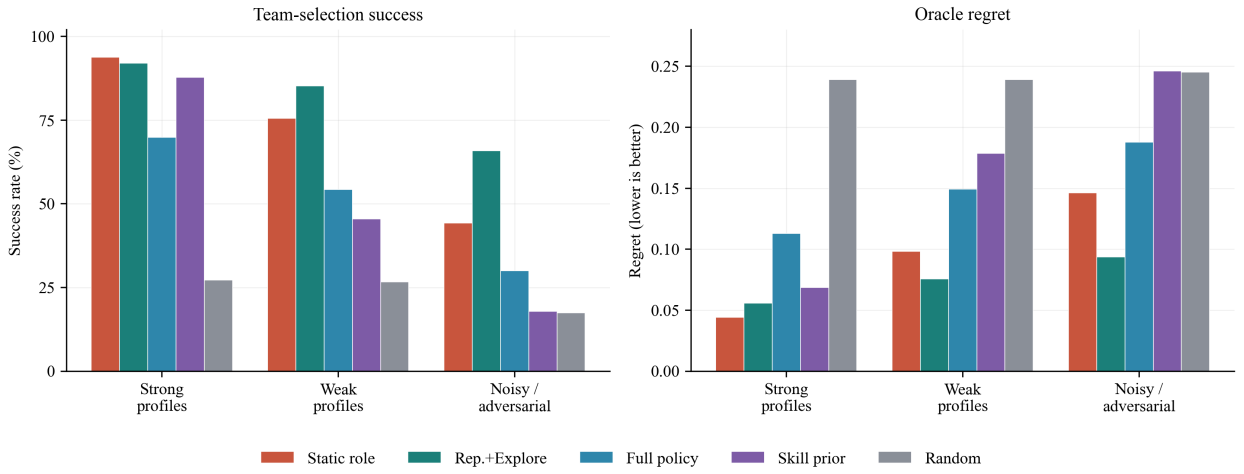


Figure 3: **Profile-sensitivity benchmark** (691,200 rows). Static role assignment is strongest when profiles are reliable; the reputation-plus-exploration policy is strongest when profiles are weak or adversarially noisy. Bars are means over 40 seeds.

5 Offline Profile-Sensitivity Benchmark

Construct. The benchmark instantiates 24 agents with hidden true per-skill abilities that define the oracle team for each subtask; the team former sees only the (degraded) visible profiles and the reputation it accrues. There are three profile regimes — *strong* (visible profiles track true skill), *weak* (profiles are uninformative), and *noisy/adversarial* (profiles are actively misleading and some agents misrepresent their skills) — 40 seeds, 360 episodes per seed/config/regime, 16 policies, and a team size of three, for 691,200 scored rows. The primary metric is oracle regret (expected-success gap to the oracle team; lower is better), with success rate, rank percentile, and selected-adversary fraction as secondary.

Result: reputation helps where profiles are unreliable. Table 2 and Fig. 3 give the picture. Under strong profiles, the fixed STATIC baseline has the lowest regret (0.044), and the seed-paired contrast favors it over REP+EXPLORE (+0.0118 regret, 95% CI $[0.0075, 0.0160]$): when declared profiles are trustworthy, a planner should trust them. Under weak profiles the ordering flips — REP+EXPLORE reaches 85.2% success at regret 0.076 versus 75.5% and 0.098 for STATIC (seed-paired -0.0224, 95% CI $[-0.0325, -0.0128]$) — and the gap widens under noisy/adversarial profiles, where REP+EXPLORE reaches 65.9% versus 44.2% (regret 0.094 vs. 0.146; seed-paired -0.0525, 95% CI $[-0.0635, -0.0424]$). The strong-profile case is a

Table 2: **Best non-oracle policy by regime** (means over 40 seeds). Oracle regret in the rightmost block is lower-is-better; the oracle reference has regret 0 at success 0.98–0.99. **Bold** marks the lower-regret policy within each regime.

Regime	Policy	Success	Oracle regret	Rank pct.	Adversary frac.
Strong	STATIC	0.938	**0.044**	0.971	0.322
	REP+EXPLORE	0.921	0.056	0.955	0.248
Weak	REP+EXPLORE	0.852	**0.076**	0.921	0.276
	STATIC	0.755	0.098	0.869	0.363
	FULL	0.543	0.149	0.741	0.307
Noisy / adv.	REP+EXPLORE	0.659	**0.094**	0.888	0.446
	STATIC	0.442	0.146	0.761	0.697
	FULL	0.300	0.188	0.639	0.587

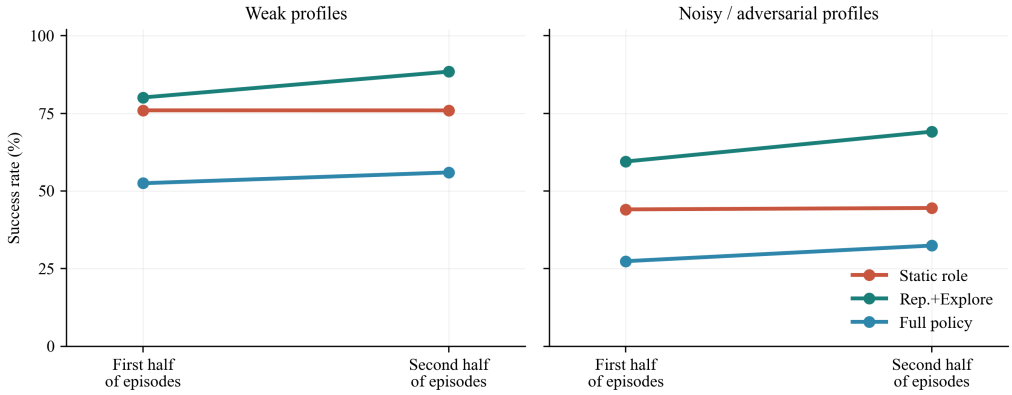


Figure 4: **Within-run learning.** Adaptive reputation policies improve from the first to the second half of episodes under weak and noisy profiles; static role assignment does not.

negative control: the system is not best in this regime, and by design it should not be.

The full objective is under-tuned, and social cohesion is the cause. The default FULL policy is markedly worse than REP+EXPLORE in every regime (weak regret 0.149 vs. 0.076, seed-paired -0.0737, 95% CI $[-0.0804, -0.0667]$, winning all 40 seeds; noisy 0.188 vs. 0.094, seed-paired -0.0939, 95% CI $[-0.1065, -0.0814]$). A dedicated diagnostic sweep over social-cohesion variants traces this to the social term: rewarding raw positive cohesion reinforces repeated co-assignment without causal evidence of complementarity, and even at low weight it underperforms turning social scoring off. We therefore do not present the default full policy as the primary result; the better-supported claim is that per-skill reputation plus exploration helps, while social cohesion needs stronger causal grounding before it should be rewarded.

Learning dynamics. Figure 4 contrasts first- and second-half episode performance. REP+EXPLORE improves within a run — weak-profile success rises from 80.1% to 88.4% and noisy/adversarial success from 59.5% to 69.1% — while STATIC is essentially flat (it cannot learn) and FULL improves more slowly from a lower base. The reputation–skill Spearman correlation for REP+EXPLORE reaches 0.68 (weak) and 0.62 (noisy), confirming that the learned posteriors track the hidden skills rather than fitting noise.

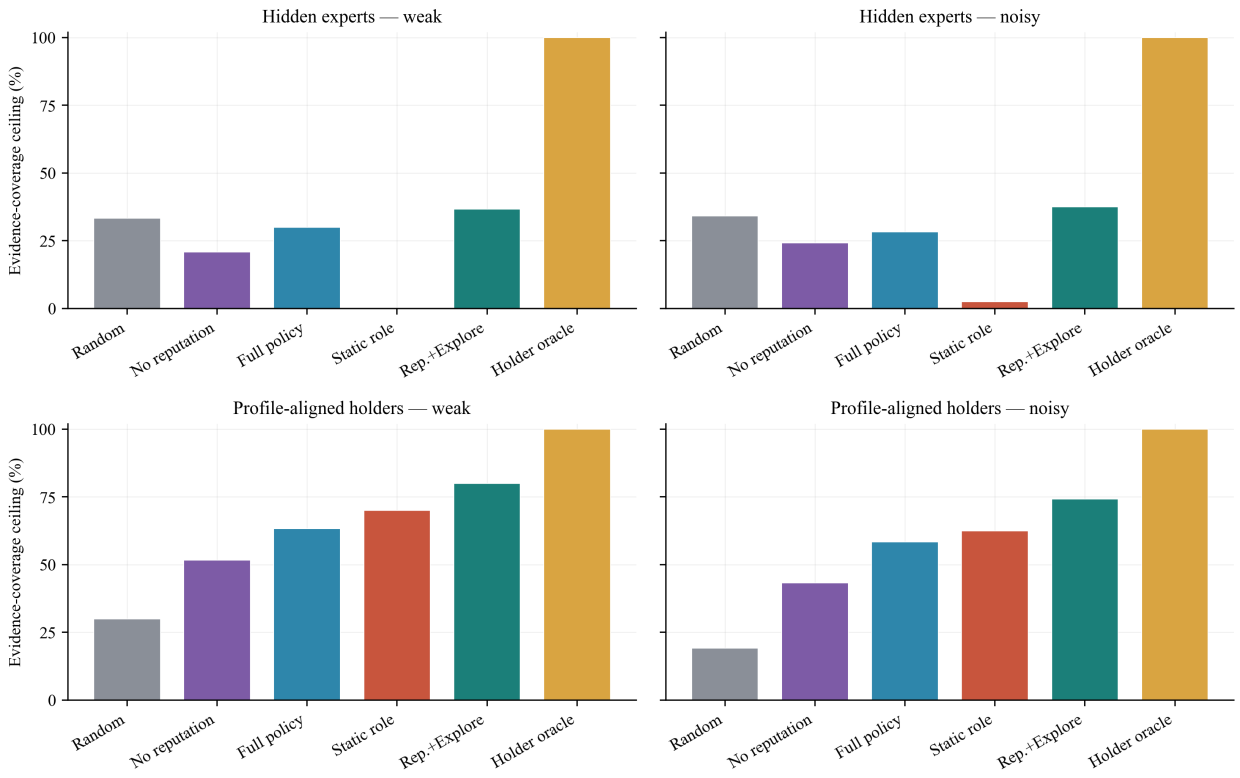


Figure 5: **Live private-evidence coverage ceiling.** Each bar is the fraction of rows whose selected team holds all required evidence. REP+EXPLORE is the strongest learned policy when holders are profile-aligned (bottom); when experts are hidden (top), every learned policy sits near RANDOM.

6 Live Private-Evidence Benchmark

The offline benchmark measures selection quality against a known oracle but uses no language model. To make team selection causally decisive with real models, we construct a private-evidence task: each required answer slot is held by exactly one agent as a private shard, the selected team executes through live model calls, and the final answer can be correct only if the team collectively holds every slot. The benchmark spans 2,880 rows across two holder policies — *hidden experts* (the holder is unrelated to the visible profile) and *profile-aligned* holders — two regimes, and six selection policies including RANDOM and a HOLDER-ORACLE that sees only the holder map (not the answer) and still issues live agent and coordinator calls.

The benchmark isolates selection. Incomplete-evidence rows have zero exact-match accuracy under both holder policies, so the task genuinely tests evidence coverage rather than guessing. Figure 5 and Table 3 report the coverage ceiling — the fraction of rows whose selected team holds all slots — and end-to-end generative accuracy. When holders are profile-aligned, REP+EXPLORE attains the highest ceiling (0.80 weak, 0.74 noisy) and generative accuracy (0.692, 0.617), closing 71.4% and 68.0% of the random-to-oracle selection gap (Fig. 6). When experts are hidden, no policy can use the visible profile to find them, and every learned policy collapses toward RANDOM: REP+EXPLORE closes only $\sim 5\%$ of the gap, and STATIC — which always picks the same profile-named agents — does worst of all (0–0.025 ceiling). This is the expected boundary of the method.

Table 3: **REP+EXPLORE in the live private-evidence benchmark.** Generative accuracy is the live coordinator's exact-match rate; the coverage ceiling is the team-selection metric; gap-closed is the fraction of the random-to-oracle coverage gap recovered. n=120 rows per cell.

Holder	Regime	Gen. accuracy	Coverage ceiling	Gap closed	Gain vs. RANDOM
Hidden expert	Weak	0.350	0.367	0.050	+0.050
Hidden expert	Noisy	0.350	0.375	0.051	+0.033
Profile-aligned	Weak	0.692	0.800	0.714	+0.442
Profile-aligned	Noisy	0.617	0.742	0.680	+0.442

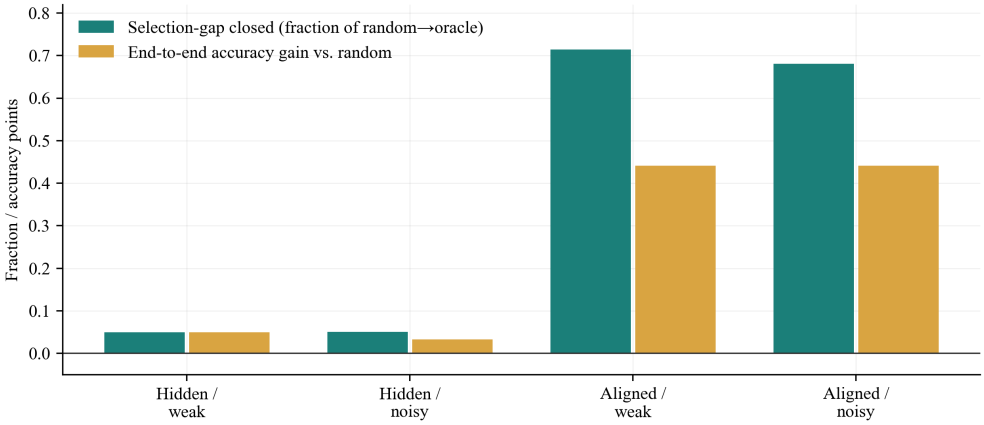


Figure 6: **Distance from random to the holder oracle.** REP+EXPLORE recovers most of the evidence-selection gap and yields a large end-to-end accuracy gain only when evidence is profile-aligned; hidden-expert discovery stays near chance.

7 Coordinator Ablation

Even with a complete-evidence team, the answer still has to be aggregated from the agents' reports, and this is a separable source of loss. We replayed all 1,440 complete-evidence rows with the team and raw agent reports held fixed, swapping the free-form coordinator for a schema-constrained JSON coordinator and for a deterministic slot-assembly procedure. The replay produced zero API errors and zero JSON parse errors, yet JSON final-answer accuracy fell to 0.586 from 0.839 for the original coordinator, while deterministic assembly from the parsed agent facts reached 0.992 (Fig. 7). The failure mode is concrete and reproducible: the schema-constrained model frequently treats an explicit absence marker such as `ALPHA=NONE` as missing evidence even when another agent supplies the real slot value. The practical implication is that, once a team covers the evidence, deterministic assembly is preferable to trusting an LLM to merge facts — aggregation loss is an interface problem, not a team-selection problem.

8 Live Backend Pilots

We ran the full backend — decomposition, ledger-driven execution, channel writers, and the structured judge — on real OpenAI calls (`gpt-4o-mini` workers, `gpt-4o` judge) with deterministic external answer checking. These pilots are small and we report them as systems gates and transfer diagnostics, not as confirmatory evidence (Table 4).

A 108-episode paired ablation over an atomic curriculum confirms channel isolation works as intended —

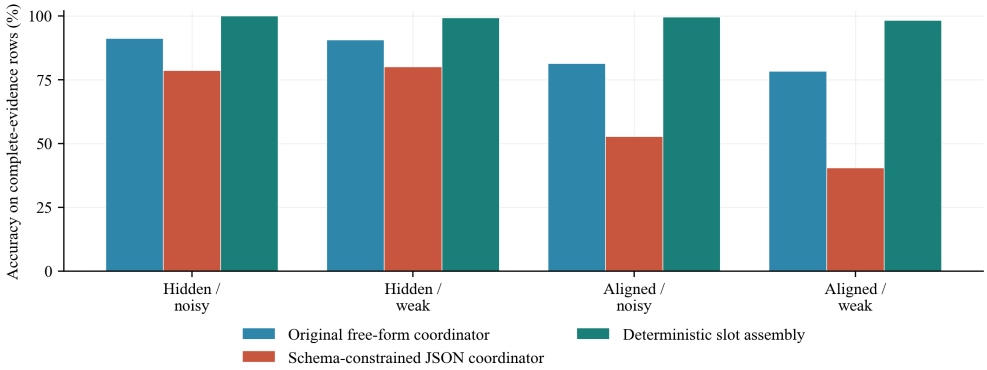


Figure 7: **Coordinator replay on complete-evidence rows** (1,440 calls). Schema enforcement removes parse errors but loses facts; deterministic slot assembly recovers them.

Table 4: **Live backend pilots** (deterministic external scoring is primary; judge score is secondary). The hard control matrix uses 8 rows per configuration.

Run	Config	Rows	Accuracy	Completion	Usable answer	Mem. / Rep. writes
API curriculum	FULL	36	0.861	1.000	—	87 / 115
	NO-MEM	36	0.833	1.000	—	0 / 123
	NO-REP	36	0.833	1.000	—	85 / 0
Hard control	FULL	8	0.750	**1.000**	**1.000**	156 / 107
	NO-REP	8	**0.875**	0.875	0.875	124 / 0
	NO-MEM	8	0.625	0.875	0.875	0 / 134
	REP+EXPLORE	8	0.625	0.875	0.750	139 / 202

NO-MEM writes zero memory entries and NO-REP writes zero reputation updates — with FULL marginally ahead on accuracy (0.861 vs. 0.833); five of six task families are saturated, so the curriculum is too easy to separate the policies. A harder 32-row control matrix is more informative and less favorable to the full system: the NO-REP control has the highest raw accuracy (0.875 vs. 0.750 for FULL) and the highest semantic accuracy when it answers (1.000), while FULL's only clean advantage is reliability — it is the one configuration that completes every row and returns a usable answer every time (Fig. 8). A separate 640-row direct-call transfer harness, which strips the backend down to policy-selected teams plus majority voting, runs without response errors but is again near-saturated on the available deterministic tasks. Live transfer of the offline selection advantage is therefore not yet demonstrated at the sample sizes and task difficulty we could afford; the offline and private-evidence benchmarks, not these pilots, carry the team-selection claim.

9 Internal Validation

The remaining experiments test whether the proposed mechanisms behave as designed on a controlled coordination substrate. They establish that the state channels and defenses are load-bearing; following the evidence ladder, we do not read them as live LLM task superiority.

Channels are load-bearing. On a leader-election coordination ladder, a leave-one-channel-out ablation at population size $N{=}32$ drops hard-score from 0.79 (all channels) to 0.74, 0.71, 0.72, and 0.65 when the profile, memory, edge, and reputation channels respectively are disabled, and to 0.51 when the judge loop is

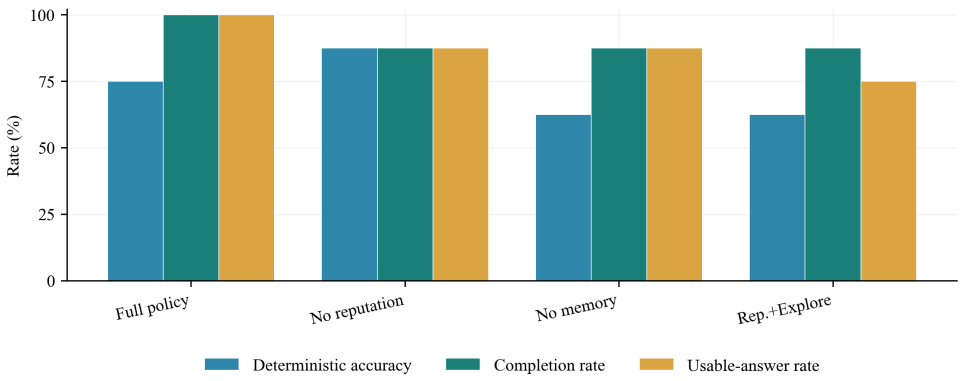
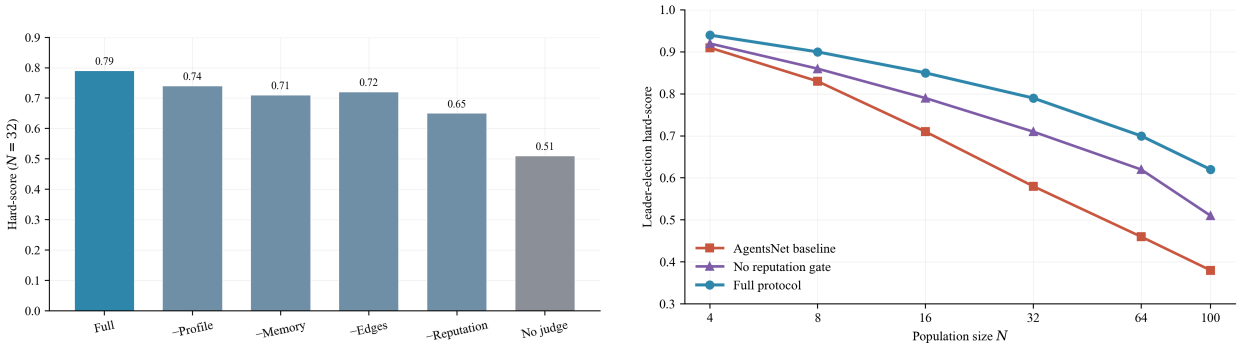


Figure 8: **Live hard-control pilot** (32 rows). The no-reputation control has the highest raw accuracy; the full system's advantage is reliability — it always completes and always returns a usable answer.



(a) Leave-one-channel-out at $N{=}32$.

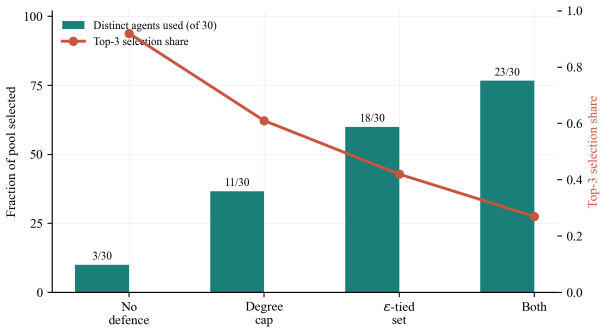
(b) Scaling with population size.

Figure 9: **Internal validation on the coordination substrate.** Removing any channel lowers hard-score; the reputation gate's contribution grows with population size.

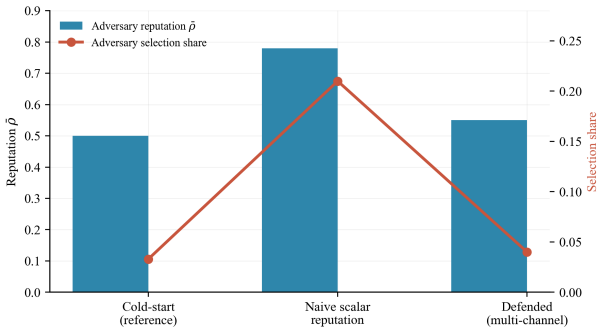
removed entirely (Fig. 9a). The reputation channel carries the largest single drop, and the further fall to 0.51 with no judge rules out the reading that judge cadence alone drives the effect. On the same ladder the full loop's advantage over the no-reputation variant grows with population size, from a small margin at $N{=}4$ to $+0.11$ at $N{=}100$ (Fig. 9b), consistent with reputation mattering more once evidence has accrued across a larger pool.

Anti-oligarchy needs both mechanisms. On a 30-agent cold-start pool over 100 episodes, no defense uses only 3 of 30 agents (top-3 selection share 0.92); the degree cap alone reaches $11/30$, the ϵ-tied set alone $18/30$, and the two together $23/30$ with top-3 share down to 0.27 (Fig. 10a). Neither mechanism is sufficient alone.

Multi-channel updates resist self-promotion. An adversarial agent that always claims full credit, pads its outputs, and rates itself maximally inflates a naive scalar reputation to 0.78 and a selection share of 0.21; under the credibility-weighted multi-channel updates its reputation stays near 0.55 and its selection share near 0.04 (Fig. 10b). Separating the actor, the peer witnesses, and the judge — and bounding each per-call increment — removes the means to game a single score.



(a) Anti-oligarchy controls.



(b) Adversarial self-promotion probe.

Figure 10: **Robustness diagnostics.** Exploration and the degree cap together prevent selection collapse; credibility-weighted updates limit adversarial reputation inflation.

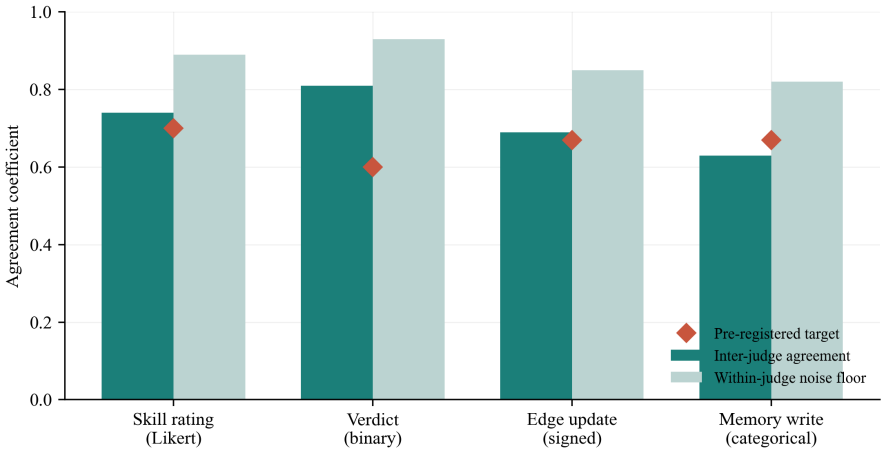


Figure 11: **Inter-judge agreement** by channel, against the pre-registered target and the within-judge noise floor. Agreement is usable but below oracle level.

Judge reliability. On a held-out replay across three judge models at temperature zero, skill-rating agreement reaches $\mathrm{ICC}(2,1)$=0.74 $[0.68, 0.79]$ and binary-verdict agreement Cohen's κ=0.81, both clearing common reliability thresholds (Shrout and Fleiss, 1979; Cicchetti, 1994), while the memory-write channel sits just below target at α=0.63 (Fig. 11). All channels stay below their within-judge noise floors, so genuine inter-judge disagreement remains, which is the reason deterministic labels and oracle ceilings are primary throughout. A cross-judge matrix shows a small self-preference effect of $+0.13$ Likert points on within-family pairs, consistent with Panickssery et al. (2024) and motivating the judge-disjoint-from-pool constraint.

10 Discussion

When does reputation help? The results converge on a conditional answer. Persistent per-skill reputation with explicit exploration improves team formation when the visible profile information is weak or actively misleading — the regimes where a planner has nothing better to fall back on. When profiles are reliable, a fixed assignment is both simpler and better, and CONOID should defer to it; our strong-profile negative control says exactly this. The advantage is also bounded by what reputation can observe: when the agent who holds critical evidence is unrelated to its advertised profile, no amount of reputation learning recovers it, and

performance returns to chance.

Why the full system is not the headline claim. The data do not support it: the default full objective is consistently worse than the social-off, exploration-heavy policy, and the cause is identifiable: rewarding raw social cohesion reinforces co-assignment cliques rather than complementarity. The supported claim is more specific than "the full system wins": two ingredients (per-skill reputation and uncertainty-driven exploration) help, while a third (naive social cohesion) needs causal grounding before it earns a place in the objective.

Two separable bottlenecks. The private-evidence and coordinator results show that "getting the answer right" decomposes into selecting a team that covers the evidence and then assembling the answer without losing facts. CONOID improves the first under weak profiles; the second is better solved deterministically than by an LLM. Treating these as one metric would have hidden both findings.

11 Limitations

The offline benchmark is synthetic by construction: it supplies hidden oracle skills so that selection can be scored exactly, and it does not measure language quality. The internal-validation suite is a set of mechanism diagnostics on a controlled coordination substrate and should not be read as broad live evidence. The live private-evidence benchmark uses real model calls but templated, exact-match tasks. The live backend pilots are small (≤ 36 rows per cell), single-provider, and runtime-heavy, and are reported as pilots. The LLM judge is reliable enough to use but not oracle-level, which is why we keep it secondary. Finally, the cross-judge analysis covers a small set of contemporary proprietary models and cannot rule out biases shared across related pretraining corpora.

12 Conclusion

CONOID casts self-improvement in an LLM-agent organization as an auditable state-update problem: a structured judge proposes bounded deltas to profile, memory, social-graph, and per-skill reputation channels, and those channels feed the next team-selection score. A large oracle-backed benchmark and a live private-evidence benchmark locate where this helps — team formation under weak or misleading profiles — and where it does not — strong profiles and hidden-expert discovery. A coordinator replay shows that answer aggregation is a separate problem better solved deterministically. The conclusion is narrow: persistent per-skill reputation with exploration improves team selection under profile uncertainty, it is not a universal win, and the path forward runs through hidden-expert discovery, cheaper live orchestration, learned team-scoring weights, and deterministic aggregation.

Reproducibility. Result artifacts (the offline summary and phase-delta tables, the private-evidence bounds, the coordinator replay, and the live pilot logs) and the figure-generation script are released with the codebase; every figure in this paper is regenerated from those artifacts.

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